

A smooth reflexive shift counterexample to the Faulkner–Huneycutt problem

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Source Problem

The source paper is Kobos and Wójcik, *On non-linear mappings preserving the semi-inner product*, arXiv:2204.06281. In Remark 4.2, after discussing the role of orthogonal complements in the characterization of property (SL), the paper quotes an old problem of Faulkner and Huneycutt:

If X is a smooth reflexive Banach space and $Y \subseteq X$ is a closed subspace isometric to X , must $Y^\perp$ be a linear subspace?

Here the source paper uses the Lumer–Giles semi-inner-product convention: in a smooth space, $x \perp y$ means $[y | x] = 0$, and

$$Y^\perp = \{x \in X : x \perp y \text{ for every } y \in Y\}.$$

The source remark spans two rendered pages:

Remark 4.2. At least in the strictly convex case it seems to be essential, that in Theorem 1.2 we refer to a subspace $V \subseteq X/Y$ of a quotient space, and not just the quotient space itself. Faulkner and Huneycutt posed the following problem in 1978 (see [4]): is it true that if X is a smooth reflexive Banach space and $Y \subseteq X$ is a closed subspace isometric to X , then $Y^\perp$ is a linear subspace? If $Y \subseteq X$ is such that X/Y is isometric to X , then by taking duals we get that $Y_0 \subseteq X^*$ is isometric to X^* , where

$$Y_0 = \{\varphi \in X^* : \varphi|_Y \equiv 0\}.$$

If X is strictly convex, then X^* is smooth. In this case, if the problem of Faulkner and Huneycutt has the positive answer, we get that $Y_0^\perp$ is a linear subspace of X^* . From here, it is now quite easy to establish that $Y^\perp$ is also a linear subspace. Thus, it is not reasonable to expect that some easy example of the space without the property (SL) can be found, where only the quotient space (and not its subspace) is used. It should be noted that the problem of Faulkner and Huneycutt has positive answer in some particular cases, but in the full generality it remains open until this day.

5. SOME CONSEQUENCES OF THEOREM 1.2

In this section we present some applications of Theorem 1.2. We begin with

Answer

Theorem 1. *Over either scalar field $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$, there is a smooth, strictly convex, reflexive Banach space X and a closed subspace $Y \subseteq X$ which is linearly isometric to X , but for which $Y^\perp$ is not a linear subspace.*

Proof Intuition

The construction is an equivalent norm on ℓ_4 . Its fourth power is a sum of fourth powers of overlapping three-coordinate linear forms. The overlap is periodic enough that the two-step shift is an exact isometry, so the shifted range is a closed copy of the whole space. But the same overlap lets us force a nonlinear orthogonality defect at the first two coordinates: the support functionals at e_1 and e_2 vanish on the shifted range, while the support functional at $e_1 + e_2$ does not vanish on e_3 .

Construction of the Space

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. For $a, b, c \in \mathbb{K}$, put

$$\Phi(a, b, c) = |a + b + c|^4 + |a - c|^4 + |b - c|^4 + |a - b|^4.$$

For a sequence $x = (x_n)_{n \geq 1}$, set $x_0 = x_{-1} = 0$ and define

$$\|x\|_X^4 = \sum_{n=1}^{\infty} \Phi(x_{n-2}, x_{n-1}, x_n).$$

Let X be the vector space $\ell_4(\mathbb{N}; \mathbb{K})$ equipped with this norm.

Lemma 1. *The norm $\|\cdot\|_X$ is equivalent to the usual ℓ_4 -norm. The resulting Banach space X is reflexive, smooth, and strictly convex.*

Proof. Define a bounded linear operator

$$B : \ell_4(\mathbb{N}; \mathbb{K}) \longrightarrow \ell_4(\mathbb{N}; \mathbb{K}^4)$$

by

$$(Bx)_n = (x_{n-2} + x_{n-1} + x_n, x_{n-2} - x_n, x_{n-1} - x_n, x_{n-2} - x_{n-1}).$$

Then $\|x\|_X = \|Bx\|_4$.

The four linear forms

$$a + b + c, \quad a - c, \quad b - c, \quad a - b$$

separate points of $\mathbb{K}^3$. Hence, by finite-dimensional norm equivalence, there are constants $m, M > 0$ such that

$$m(|a|^4 + |b|^4 + |c|^4) \leq \Phi(a, b, c) \leq M(|a|^4 + |b|^4 + |c|^4)$$

for all $a, b, c \in \mathbb{K}$. Summing the lower bound over n gives

$$\|x\|_X^4 \geq m \sum_{n=1}^{\infty} |x_n|^4,$$

while the upper bound gives

$$\|x\|_X^4 \leq 3M \sum_{n=1}^{\infty} |x_n|^4.$$

Thus $\|\cdot\|_X$ is equivalent to the usual ℓ_4 -norm, so X is reflexive.

Moreover B is an isometric embedding from X into $\ell_4(\mathbb{N}; \mathbb{K}^4)$. Since ℓ_4 is smooth and strictly convex, its closed subspace $B(X)$ is smooth and strictly convex. Therefore X has the same two properties. $\square$

The Isometric Copy

Let $S : X \rightarrow X$ be the two-step shift

$$(Sx)_1 = (Sx)_2 = 0, \quad (Sx)_n = x_{n-2} \quad (n \geq 3).$$

For every $n \geq 1$, the $n + 2$ -nd three-coordinate window of Sx is the n -th three-coordinate window of x , and the first two windows of Sx are zero. Therefore

$$\|Sx\|_X^4 = \sum_{n=1}^{\infty} \Phi((Sx)_{n-2}, (Sx)_{n-1}, (Sx)_n) = \sum_{n=1}^{\infty} \Phi(x_{n-2}, x_{n-1}, x_n) = \|x\|_X^4.$$

Thus S is a linear isometry. Its range

$$Y = S(X) = \{y \in X : y_1 = y_2 = 0\}$$

is closed and isometric to X .

Support Functional Computation

For real-coordinate vectors x , the support functional can be written explicitly. If $x \neq 0$, then

$$\|x\|_X^3 \varphi_x(h) = \sum_{n=1}^{\infty} \sum_{j=1}^4 L_j(h_{n-2}, h_{n-1}, h_n) L_j(x_{n-2}, x_{n-1}, x_n)^3,$$

where

$$L_1(a, b, c) = a + b + c, \quad L_2(a, b, c) = a - c, \quad L_3(a, b, c) = b - c, \quad L_4(a, b, c) = a - b.$$

This is just the pullback through B of the usual support functional of ℓ_4 : for $z \neq 0$ in ℓ_4 , that functional is $\|z\|_4^{-3} \sum_i u_i |z_i|^2 \overline{z_i}$ on a test vector u . For $\mathbb{K} = \mathbb{C}$ the same formula has $L_j(x)^3$ replaced by $|L_j(x)|^2 \overline{L_j(x)}$; for the real-coordinate witnesses used below these expressions agree.

Let

$$\begin{aligned} A(a, b, c) &= (a + b + c)^3 + (a - c)^3 + (a - b)^3, \\ B(a, b, c) &= (a + b + c)^3 + (b - c)^3 - (a - b)^3, \\ C(a, b, c) &= (a + b + c)^3 - (a - c)^3 - (b - c)^3. \end{aligned}$$

For a real-coordinate x , the coefficient of h_k in $\|x\|_X^3 \varphi_x(h)$ is

$$\Gamma_k(x) = C(x_{k-2}, x_{k-1}, x_k) + B(x_{k-1}, x_k, x_{k+1}) + A(x_k, x_{k+1}, x_{k+2}).$$

Since $Y = \overline{\text{span}\{e_k : k \geq 3\}}$, a vector x belongs to $Y^\perp$ exactly when $\Gamma_k(x) = 0$ for every $k \geq 3$.

Nonlinearity of $Y^\perp$

First take $x = e_1$. For $k = 3$,

$$\Gamma_3(e_1) = C(1, 0, 0) + B(0, 0, 0) + A(0, 0, 0) = 0,$$

and for $k \geq 4$ every contributing window is zero. Hence $e_1 \in Y^\perp$.

Next take $x = e_2$. We have

$$\Gamma_3(e_2) = C(0, 1, 0) + B(1, 0, 0) + A(0, 0, 0) = 0$$

and

$$\Gamma_4(e_2) = C(1, 0, 0) + B(0, 0, 0) + A(0, 0, 0) = 0,$$

while $\Gamma_k(e_2) = 0$ for $k \geq 5$. Hence $e_2 \in Y^\perp$.

Finally take $x = e_1 + e_2$. In the direction $e_3 \in Y$,

$$\Gamma_3(e_1 + e_2) = C(1, 1, 0) + B(1, 0, 0) + A(0, 0, 0) = (8 - 1 - 1) + (1 - 1) = 6 \neq 0.$$

Therefore $\varphi_{e_1+e_2}(e_3) \neq 0$, so

$$e_1 + e_2 \notin Y^\perp.$$

We have shown that $e_1, e_2 \in Y^\perp$ but $e_1 + e_2 \notin Y^\perp$. Thus $Y^\perp$ is not a linear subspace, completing the proof of Theorem 1. $\square$

Verification Notes

The script

`code/verify_counterexample.py`

checks the displayed support-functional coefficients by exact rational arithmetic. It also checks on finite samples that the two-step shift preserves the fourth power of the norm. The command used was:

`python3 code/verify_counterexample.py.`

The script is not needed for the proof; it is a reproducibility check for the finite algebra in the witness computation.

Limitations and Novelty Check

This packet gives a counterexample to the smooth reflexive orthogonal complement problem quoted in arXiv:2204.06281. It does not assert anything about variants with extra hypotheses beyond smoothness, reflexivity, and an isometric closed copy of the whole space.

Before promotion, the run’s cheap indexes were searched for arXiv:2204.06281, the phrases “Faulkner Huneycutt”, “smooth reflexive”, “ $Y^\perp$ linear”, and nearby semi-inner-product terms. No exact prior packet or attempt was found. Local/source-corpus searches found related isometric-range and one-complemented-subspace literature, including Pelczar-Barwacz arXiv:2309.15261 and discussion in arXiv:2407.15112, but not the smooth semi-inner-product counterexample above. Web/arXiv spot checks on 2026-06-30 for the same exact phrases found no later paper explicitly resolving the quoted problem.

Human Review Recommendation

The main review point is the support-functional formula for the pullback ℓ_4 -norm, together with the source paper’s convention $x \perp y \Leftrightarrow [y \mid x] = 0$. Once that convention is fixed, the failure of linearity is the explicit calculation

$$e_1, e_2 \in Y^\perp, \quad e_1 + e_2 \notin Y^\perp.$$

References

- [1] T. Kobos and P. Wójcik, *On non-linear mappings preserving the semi-inner product*, arXiv:2204.06281.
- [2] G. D. Faulkner and J. E. Huneycutt Jr., *Orthogonal decomposition of isometries in a Banach space*, Proc. Amer. Math. Soc. **69** (1978), 125–128.